1. Sol

* Electric field lines show the strength and direction of electric force around a charge. Around a positive charge, lines move outward, while around a around a negative charge, lines move inward.

* When two opposite charges are near each other, field lines connect from positive to negative, shawing attractive force. for like charges, field lines spread apart, showing repulsive force.

* Therefore, electric feild line patterns help us visually understand the direction, strength, attraction and repulsion of electric forces between charges.

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2. * Electric field representation is better for systems with multiple charges because it gives the effect of all charges at a point directly

* It is simple and more efficient than calculating seperate force vectors for every charge interaction

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3. * A capacitor charges and discharges to create time delay in electronic circuit.

* In a RC circuit, the timing depends on the values of resistance and capacitance

* stable dielectric materials keep capacitance constant, giving accurate and consistent timing

4. * Proper capacitor selection improves device reliability by ensuring safe operation under different voltages and preventing break down or failure

* Engineers consider dielectric strength, durability, performance and cost to achieve safe, efficient and long lasting electronic devices.

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5. * Identify the positions and magnitudes of all fixed charges in the system.

* calculate the electric feild produced by each charge at the required point using ~~...~~ Coulomb's law.

* Add all individual electric feilds vectorically to obtain the net feild intensity accurately.

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6. * Discrete summation is suitable for a small number of charges, but calculations becomes complex and time consuming as the number of charges increases.

* For a large system, continuous charge distribution is more efficient because integration gives the overall ~~of~~ electric field in a simple and symmatic way

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7. * Identify the required equivalent capacitance and the values of available capacitors

* use parallel combination to increase capacitance ($C_1 = C_2 + C_3 + C_3$) and series combination to decrease capacitance ($\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots$)

* combine capacitors step by step and verify the final equivalent capacitance to ensure the correct arrangement is obtained

8. * Increasing capacitance improves energy storage and voltage stability, which increases system efficiency and performance.

* However, larger capacitance increases size, cost, and heat generation, reducing scalability and making system design more difficult.

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9. *According to Gauss's law, the total electric flux through a closed surface is equal to the enclosed charge divided by permitivity of free space

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{enclosed}}{\epsilon_0}$$

For a charged spherical shell,

$$E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$$

$$\cancel{A\ 4\pi r}\ A = 4\pi r^2$$

electric flux is $\Phi = E \times A$

$$\Phi = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \times 4\pi r^2$$

$$\boxed{\Phi = \frac{Q}{\epsilon_0}}$$

*Thus, the radius r cancels out, showing that the electric flux remains constant for any guassian surface enclosing the same charge.

10. * In electrostatic equilibrium, the electric field inside a grounded conducting chamber should be zero.

* Small gaps, joints or cracks in the conducting walls allow external fields to enter the chamber.

* Poor conductivity of the material can prevent proper redistribution of charges

* Improper grounding may leave induced charges on the chamber surface, causing internal electric fields.

* Hence, imperfections in the conductors and faulty grounding disturb electrostatic sheilding and produce non-zero electric field inside the chamber

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11. Electric flux through a flat surface in a uniform electric feild is given by:

$$\Phi = \vec{E} \cdot \vec{A} = EA\cos\theta$$

E = magnitude of electric field
A = area of the surface
Φ = angle between electric field and the area vector

* The flucating values occur because the angle is measure incorrectly with the surface instead of normal. This leads to wrong projection of area and incorrect use of $EA\sin\theta$ instead of $EA\cos\theta$

* Flux is maximum when the field is prependicular to the surface and zero when parallel

* Hence, the error is due to misunderstanding of surface orientation and flux direction

12. * The torque on an electric dipole in a uniform electric field

$$\tau = PE\sin\theta$$

Where
$p = qd$ is the dipole moment
E = Electric field strength
θ = angle between dipole axis and electric field

* The torque is depend on angular position:
i) Torque is maximum at $\theta = 90^\circ$
ii) Torque is ~~bein~~ zero at $\theta = 0^\circ$ or 180°

* Thus, the dipole tends to rotate and align itself parallel to the electric field. Greater angular displacement produces greater rotational tendency up to 90°

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13. * To minimize energy consumption while maintaining measurable potential difference, the charges should be arranged with moderate and uniform spacing to create a nearly uniform electric feild.

* The relation between electric field & potential is

$$E = -\frac{dV}{dx}$$

* work done is

$$W = q\,\Delta V$$

* If charges are placed too close, the electric field becomes very strong, increasing work and energy usage.
* If placed too far apart, the potential differential becomes too small for sensing
* Hence, an optimal configuration uses evenly spaced charges to produce a stable electric feild with sufficient potential difference and reduced energy expenditure

14.

* Resistance of a wire is $R = \rho \frac{L}{A}$

When the wire is doubled, $R' = 2R$

* If temperature increases, resistivity ρ also increases for copper, so resistance becomes greater than $2R$

* using ohm's law, $V = IR$ $\Rightarrow I = \frac{V}{R}$

current inversly decreases with resistance

* Drift velocity is $v_d = \frac{I}{nAe}$

since, current decreases, drift velocity also decreases

* Thus, doubling the length increases resistance, temperature rise further increases resistivity and both effects reduce current and drift velocity under constant voltage condition

15. * Electric field and potential are related by

$$E = -\frac{dV}{dx}$$

* Uneven spacing of equipotential surfaces means the electric field is non-unifor. The inconsistency is caused by incorrent modeling of discrete and continuous charge distributions and neglect of superposition effects.

* This produces wrong electric field values, causing charged particles to deviate from expected paths and give incorrect estimation of stored electrical energy

16. * Resistance of a transmission wire is

$$R = \rho \frac{L}{A}$$

* Low resistivity materials such as copper or aluminium reduce resistance and power losss.

* Increasing wire thickness increases area A which further decrease resistance and improves current carrying capacitor

* Power loss is $P = I^2R$. so, lower ~~of~~ resistance gives better efficiency.

<u>Trade-offs</u>

i) Copper - better conductivity and strong but expensive and heavy

ii) Aluminium → cheaper and lighter, but conduct less

iii) Thicker wire → lower power loss, but higher cost and weight

iv) Higher temperature → increases resistance and reduces efficiency

* Hence, a lower resistance conductor with moderate thickness gives the best balance of efficiency, cost and realiability for power transmission

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7. In the analogy:

* Each person represents a magnetic domain
* Random facing directions represent an unmagnetized material.
* Music represents the external magnetic feild
* People aligning in one direction represents magnetization

When the music stops, thermal motion and internal disturbances cause domains to return to random orientations, so the material loses its magnetization

18. * The hidden assumption is that magnetism occurs only due to visible current in conductors.

* Magnetism also comes from electron spin and orbital motion inside atoms. Alignment of these atomic magnetic moments produces magnetism even without visible current flow.

19. To obtain the required equivalent capacitance:

*use parallel combination to increase capacitance

$$C_p = C_1 + C_2 + C_3 + \cdots$$

*use series combination to decrease capacitance

$$\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots$$

* The engineer should first compare the required capacitance with available values, then choose suitable series or parallel arrangements while considering space, voltage rating and circuit performance constraints.

20. * Use a capacitor with sufficient capacitance and suitable voltage rating to store and supply charge during sudden load changes

* choose compact, low-cost capacitors that provide fast response and good stability while fitting the design space constraint

Remaining questions will be added soon